Minsoo Cheong

Integrated M.S./Ph.D. Student in Computer Science and Engineering

Computing and Memory Architecture Laboratory (CMALab) | Seoul National University

Seoul, Republic of Korea | icycle0409@snu.ac.kr | Website | Google Scholar | GitHub | LinkedIn

I am a second-year integrated M.S./Ph.D. student with 2+ years of industry experience as a software engineer. I work broadly on efficient machine learning systems, with recent research spanning language model inference, KV-cache optimization, diffusion language models, and low-bit quantization.

EDUCATION

Seoul National University · Seoul, Republic of Korea Sep 2025 - Present

Integrated M.S./Ph.D. in Computer Science and Engineering

Advisor: Prof. Sungjoo Yoo | Computing and Memory Architecture Laboratory (CMALab)

Seoul National University · Seoul, Republic of Korea Mar 2019 - Aug 2025

B.S. in Computer Science and Engineering, *Cum Laude*

PUBLICATIONS

[NeurIPS 2026] EntropyCache: Decoded Token Entropy Guided KV Caching for Diffusion Language Models **[Paper]** **[Code]**
Minsoo Cheong, Donghyun Son, Woosang Lim, and Sungjoo Yoo

[Preprint] LoopVQ: Sub 1-bit KV Cache Quantization for Looped Transformers **[Project Page]** **[Paper]** **[Code]**
Minsoo Cheong and Sungjoo Yoo

[Preprint] Tailoring the Quantization Space for 1-Bit KV Cache Compression **[Project Page]** **[Paper]** **[Code]**
Minsoo Cheong*, Donghyun Son*, and Sungjoo Yoo | ** Equal contribution.*

[Preprint] KV-Rescue: Recovering Reasoning Language Model KV Eviction Loss via Stepwise Interleaving **[Paper]**
Minsoo Cheong, Woosang Lim, Vincent-Daniel Yun, and Sungjoo Yoo

[Preprint] Output-Aware Rotation for INT2 KV-Cache Quantization **[Paper]** **[Code]**
Vincent-Daniel Yun, Woosang Lim, **Minsoo Cheong**, Sunwoo Lee, Murali Annavaram, Sai Praneeth Karimireddy, and Sungjoo Yoo

PRESENTATIONS AND ACADEMIC SERVICE

EntropyCache. Poster presentations at NeurIPS 2026 and the ICML 2026 AdaptFM Workshop; talk at the Samsung MX Research Project Exchange Seminar, Jul 2026.

Reviewer, NeurIPS 2026 Workshop on On-Device Intelligence: Foundation Models under Real-World Constraints (ODI).

OPEN SOURCE CONTRIBUTIONS

llama.cpp | Collaborator Mar 2024 - Present

Contributions to stochastic speculative decoding, the retrieval example, and EXAONE model support in the widely used C/C++ LLM inference runtime. **[Contributions]**

Spring Framework, Armeria, gRPC & Protocol Buffers | Contributor

Contributions spanning Java/Kotlin frameworks, RPC infrastructure, and serialization tooling. **[Contributions]**

krotoDC Open-source project

A protoc plugin that generates Kotlin data classes and coroutine-friendly gRPC services and stubs, making Protocol Buffers and gRPC more idiomatic in Kotlin; distributed through Maven Central with 5,000+ monthly downloads. **[GitHub]**

INDUSTRY EXPERIENCE

NAVER Corporation · Research Engineer Intern, LLM Solution Team Jun 2024 - Aug 2024

Built a Kubeflow pipeline for automated LLM inference benchmarking across hardware, batching, sequence length, quantization, and speculative decoding; implemented llama.cpp support for an internal language model.

Mathpresso · Backend Software Engineer, Content Platform Team Jun 2022 - Jun 2023

Built Kotlin/Go content-production and distribution systems and gRPC APIs; contributed to three release cycles of a digital school-exam product that achieved a 60-fold increase in sales.

Bucketplace · Backend Software Engineer, Commerce Service Team May 2021 - Jun 2022

Developed commerce backend services using Kotlin and Spring Boot, including bulk product management and microservice data aggregation.

TECHNICAL SKILLS

Machine Learning & Systems: PyTorch, CUDA, Hugging Face Transformers, LLM inference, KV caching, model efficiency

Languages & Infrastructure: Python, C++, Kotlin, Java; Git, Protocol Buffers, gRPC, Spring Boot, Project Reactor, Kotlin Coroutines